AAPL Next-Day Volatility Forecast

Decision brief for portfolio risk monitoring

EXECUTIVE SUMMARY

Use as a monitoring baseline, not an automated risk control. The model adds useful next-day ranking information but can miss the size of genuine shocks.

- Average held-out miss is 0.93 percentage points of daily absolute return.
- Error rises about 31% when current 21-day volatility is elevated.
- Robust Huber loss improves typical-day MAE, but slightly worsens RMSE and does not solve tail misses.

RECOMMENDATION

Adopt for daily triage with a human escalation rule. Do not use the point estimate to automate hedges, position limits, or trade execution.

DECISION METRICS

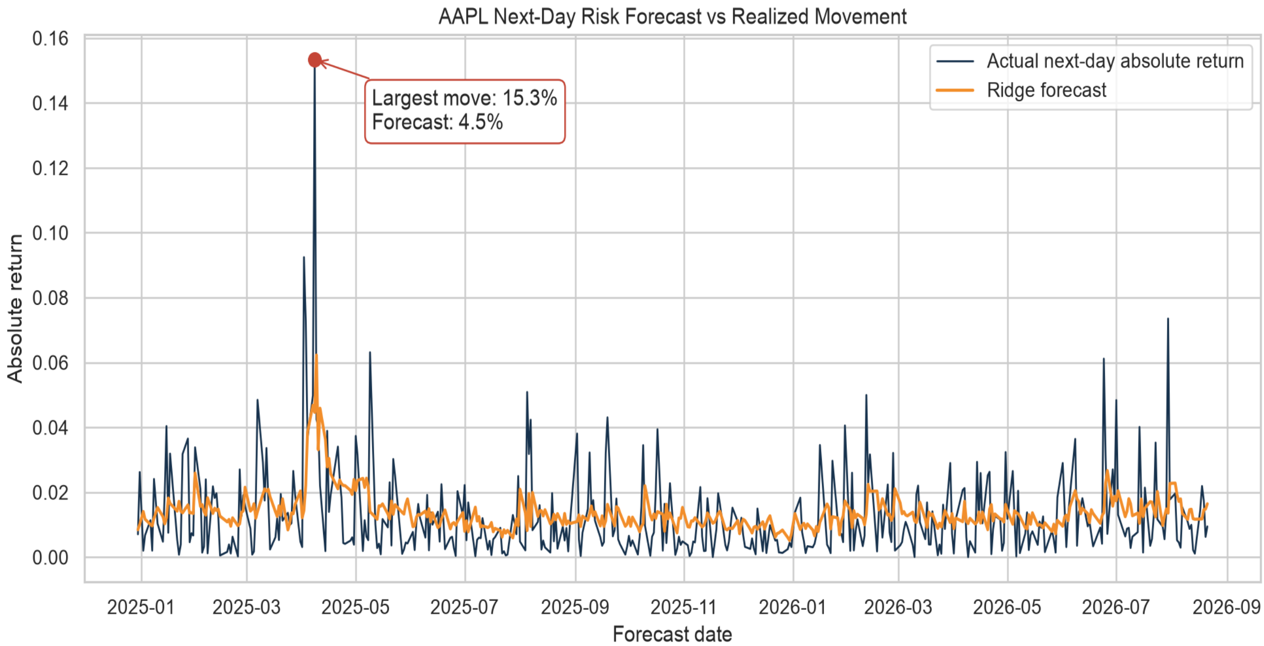
RIDGE TEST MAE	RIDGE TEST R2	BLOCK 95% CI	ELEVATED MAE
0.93 pp	**0.13**	**0.83-1.05 pp**	**1.06 pp**
Typical absolute miss	Out-of-sample variance explained	Average-MAE uncertainty	Harder operating regime

Evaluation window: 2024-12-30 to 2026-08-21 | 412 daily forecasts | Target: next-session absolute adjusted-close return

1 | FORECAST BEHAVIOR

The model tracks changing risk levels, but compresses the largest shocks

The orange forecast responds to recent volatility and usually stays near the center of realized outcomes. That stability helps routine monitoring, but it also smooths the events a risk team cares about most.



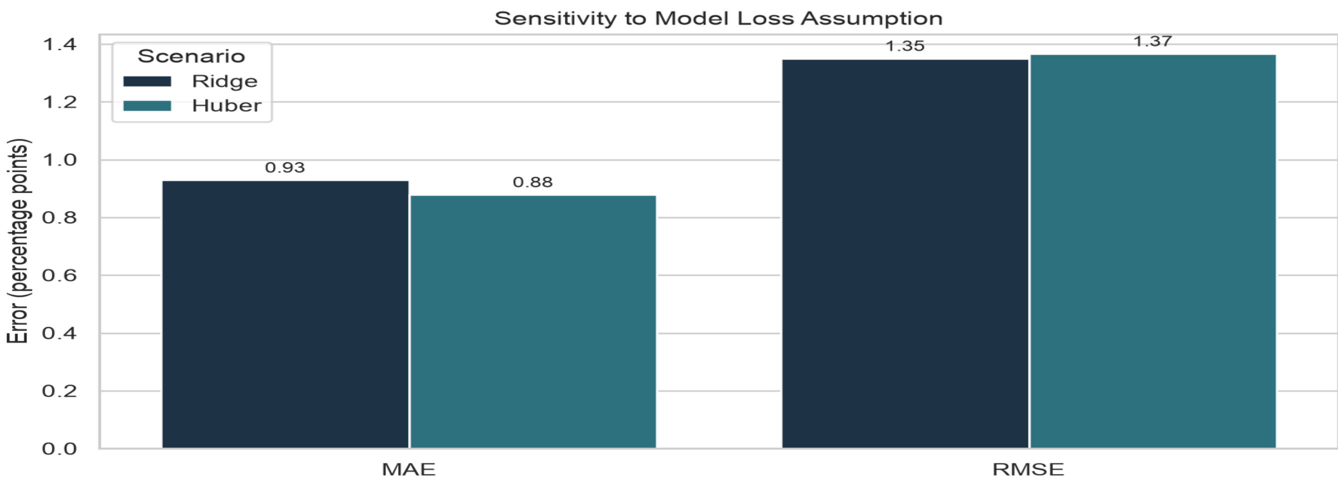
WHAT THE CHART SHOWS

On forecast date 2025-04-08, the next-day move was 15.33% while Ridge forecast 4.46%. The model identified higher risk but understated its magnitude.

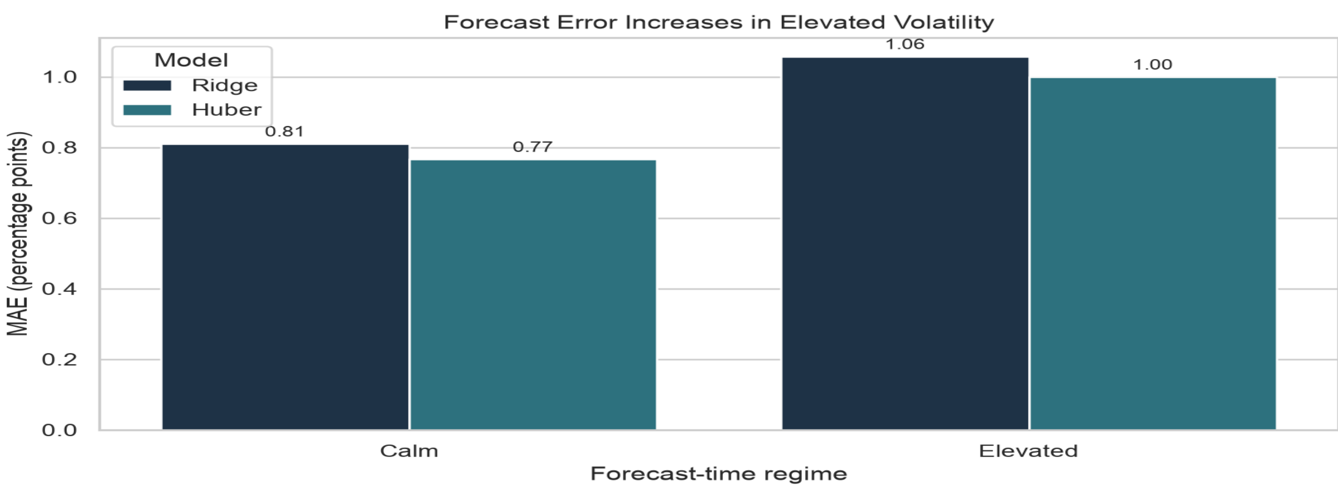
Assumption / limitation: historical lag, range, volume, and rolling-volatility relationships are assumed to carry forward. A regime break or event outside the training experience can invalidate that assumption.

The preferred model depends on the cost of typical misses versus tail misses

Two alternate assumptions are tested on the same held-out dates: squared-loss Ridge and robust-loss Huber. The regime split uses only current 21-day volatility and a threshold estimated from training data.



Huber lowers MAE by 5.4%, but raises RMSE by 1.2%. It improves the typical day while giving up a small amount of large-error performance. This is a stakeholder loss-function decision, not a universal win.



Ridge MAE increases from 0.81 pp in calm conditions to 1.06 pp when 21-day volatility exceeds 1.60%. Overall MAE hides this operating-regime gap.

3 | ASSUMPTIONS, RISKS, AND ACTION

What this means for the risk analyst

RECOMMENDED OPERATING RULE

Use the score to rank next-day attention. When current 21-day daily volatility is above 1.60%, elevate the review and do not rely on the point forecast alone.

UNCERTAINTY IN AVERAGE TEST MAE

Method	95% interval (percentage points)	What it assumes
Gaussian	0.83 - 1.02	Independent errors; normal mean approximation
IID bootstrap	0.84 - 1.03	Observed dates are exchangeable
5-day block	0.83 - 1.05	Short local dependence is retained

SENSITIVITY SUMMARY

Assumption / scenario	Change from baseline	Decision meaning
Huber robust loss	MAE -5.4%; RMSE +1.2%	Better typical day; no tail-risk cure
Elevated volatility	Ridge MAE +30.5%	Require human review / wider buffer
5-day dependence	CI width +21% vs IID	Use block interval for planning

KEY ASSUMPTIONS AND RISKS

- Data and feature relationships remain similar to the 2020-2026 history.
- The close-to-next-close target is a risk proxy, not a portfolio loss forecast or causal estimate.
- Average-error intervals do not cover refitting uncertainty, data revisions, or an unseen crisis regime.
- Correlated rolling features and changing volatility regimes can destabilize coefficients and calibration.

NEXT CONTROL STEPS

1. Run rolling-origin refits and publish tail MAE and high-volatility recall. 2. Add block-bootstrap or conformal prediction intervals. 3. Require human review above the volatility threshold. 4. Do not automate hedges or exposure limits from this prototype.